

E-Commerce Customer Behavior Analytics: A Data-Driven Business Intelligence Approach Using Python, MySQL, and Power BI

1. Project Overview

The **E-Commerce Customer Behavior Analytics** project focuses on analyzing customer shopping behavior using a transactional dataset containing **3,900 customer purchase records** across multiple product categories. The project aims to transform raw transactional data into meaningful business insights by leveraging **Python**, **MySQL**, and **Power BI** throughout the complete data analytics lifecycle.

The analysis examines customer demographics, purchasing patterns, product preferences, subscription behavior, shipping methods, and the impact of discounts on customer spending. Data preprocessing and feature engineering were performed using Python, followed by SQL-based business analysis in MySQL to answer key business questions. Finally, an interactive Power BI dashboard was developed to visualize key performance indicators (KPIs) and analytical findings.

The insights generated from this project enable businesses to better understand customer behavior, identify high-value customer segments, optimize promotional strategies, improve customer retention, and support data-driven decision-making. This project demonstrates an end-to-end business intelligence solution by integrating data cleaning, database management, SQL analytics, and interactive dashboard reporting into a unified workflow.

2. Dataset Summary

The project utilizes an **E-Commerce Customer Shopping Behavior** dataset containing customer demographic information, purchase details, and shopping behavior attributes. The dataset was used to analyze customer purchasing patterns, revenue trends, and product preferences.

Dataset Overview

Attribute	Description
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Dataset Name	E-Commerce Customer Shopping Behavior Dataset
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Total Records	3,900
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Total Features	19 Columns
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Data Type	Structured Transactional Data
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Source	Kaggle (Customer Shopping Behavior Dataset)
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Attribute **Description**

Missing Values 37 missing values in the **Review Rating** column

Target Purpose Customer Behavior Analysis and Business Intelligence

3. Exploratory Data Analysis (EDA) and Data Preprocessing Using Python

The Exploratory Data Analysis (EDA) phase was performed using **Python** in **Jupyter Notebook** to understand the dataset, improve data quality, and prepare it for further analysis. Various preprocessing techniques were applied to ensure the dataset was clean, consistent, and suitable for SQL analysis and Power BI visualization.

- **Data Loading:** Imported the dataset using pandas.

```
# Loading the dataset using pandas

import pandas as pd

df = pd.read_csv('customer_shopping_behavior.csv')

df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Visa
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	MasterCard
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Visa
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	MasterCard
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	Visa

- **Initial Exploration:** Used `df.info()` to check structure and `df.describe()` for summary statistics.

```
# Display dataset information
df.info()

# Generate summary statistics
df.describe()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   str
3   Item Purchased                       3900 non-null   str
4   Category                             3900 non-null   str
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   str
7   Size                                 3900 non-null   str
8   Color                                3900 non-null   str
9   Season                               3900 non-null   str
10  Review Rating                        3863 non-null   float64
11  Subscription Status                  3900 non-null   str
12  Shipping Type                        3900 non-null   str
13  Discount Applied                     3900 non-null   str
14  Promo Code Used                      3900 non-null   str
15  Previous Purchases                   3900 non-null   int64
16  Payment Method                       3900 non-null   str
17  Frequency of Purchases                3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 835.9 KB

```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

- **Missing Data Handling:** The dataset was examined for missing values using the `isnull()` function. It was observed that the `review_rating` column contained 37 missing values. To maintain data quality and avoid losing valuable records, the missing values were imputed using the median review rating. Median imputation was selected because it is less affected by outliers and preserves the overall distribution of the data.

```
# Check missing values
```

```
df.isnull().sum()
```

```
# Fill missing values with median
```

```
df['review_rating'].fillna(df['review_rating'].median(),
inplace=True)
```

```
# Verify missing values
```

```
df.isnull().sum()
```

Before Handling Missing Values-

Size	0
Color	0
Season	0
Review Rating	37
Subscription Status	0

After Handling Missing Values-

Size	0
Color	0
Season	0
Review Rating	0
Subscription Status	0

- **Column Standardization:** Renamed columns to snake case for better readability and documentation.

- **Feature Engineering:** Feature engineering was performed to create new variables that enhance the analytical capabilities of the dataset. These engineered features provide meaningful customer segmentation and enable more detailed business analysis.

- **Creating the age_group Column:** Customers were categorized into four age groups using the `qcut()` function based on the distribution of customer ages. This feature helps analyze purchasing behavior across different age segments.

```
#create a new column age_group  
  
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']  
  
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)  
  
df[['age', 'age_group']].head(15)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged

- **Creating the purchase_frequency_days Column:** The categorical purchase frequency values were converted into numerical day values to simplify quantitative analysis and improve visualization in Power BI.

```
purchase_frequency_mapping = {  
    'Weekly': 7,  
    'Bi-Weekly': 14,  
    'Fortnightly': 15,  
    'Monthly': 30,  
    'Quarterly': 90,  
    'Annually': 365  
}  
  
df['purchase_frequency_days'] =  
df['frequency_of_purchases'].map(  
    purchase_frequency_mapping  
)
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly

- **Data Consistency Check:** To ensure the dataset contained only relevant and non-redundant information, a consistency check was performed on the discount_applied and promo_code_used columns. The analysis showed that both columns contained identical information, making one of them redundant. Therefore, the promo_code_used column was removed to simplify the dataset and eliminate duplicate information.
- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

```

from sqlalchemy import create_engine

from urllib.parse import quote_plus

user = "root1"

password = quote_plus("XXXXXXXXXXXX")

host = "localhost"

port = "3306"

database = "customer_behavior"

engine = create_engine(

    f"mysql+pymysql://{user}:{password}@{host}:{port}/{database}"

)

table_name = "customer"

df.to_sql(

    table_name,

    engine,

    if_exists="replace",

    index=False

)

print("Data uploaded successfully!")

pd.read_sql("SELECT * FROM customer LIMIT 5;", engine)

```

Data uploaded successfully!

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status	shipping_type	d
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

- **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender	revenue
▶	Male	157890
	Female	75191

- **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62

- **Top 5 Products by Rating** – Found products with the highest average review ratings.

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

- **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

shipping_type	ROUND(AVG(purchase_amount),2)
Express	60.48
Standard	58.46

- **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

- **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

- **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	Number of Customers
Loyal	3116
Returning	701
New	83

- **Top 3 Products per Category** – Listed the most purchased products within each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

- **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
Yes	958
No	2518

- **Revenue by Age Group** – Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

- **Customer Lifetime Value (CLV)**-- Calculated the total purchase value generated by each customer throughout the dataset to identify high-value customers.

customer_id	total_orders	customer_lifetime_value	average_order_value
205	1	100	100.00
456	1	100	100.00
43	1	100	100.00
194	1	100	100.00
582	1	100	100.00
96	1	100	100.00
249	1	100	100.00
244	1	100	100.00
519	1	100	100.00
616	1	100	100.00

- **Revenue by Location**-- Compared revenue contribution across different customer locations to identify the highest-performing regions.

location	total_customers	total_revenue	average_purchase
Montana	96	5784	60.25
Illinois	92	5617	61.05
California	95	5605	59.00
Idaho	93	5587	60.08
Nevada	87	5514	60.08
Alabama	89	5261	59.11
New York	87	5257	60.43
North Dakota	83	5220	62.89
West Virginia	81	5174	63.88
Nebraska	87	5172	59.45
New Mexico	81	5014	61.90
Minnesota	88	4977	56.56
Pennsylvania	74	4926	66.57

5. Power BI Dashboard

An interactive **Power BI** dashboard was developed to present business insights visually.

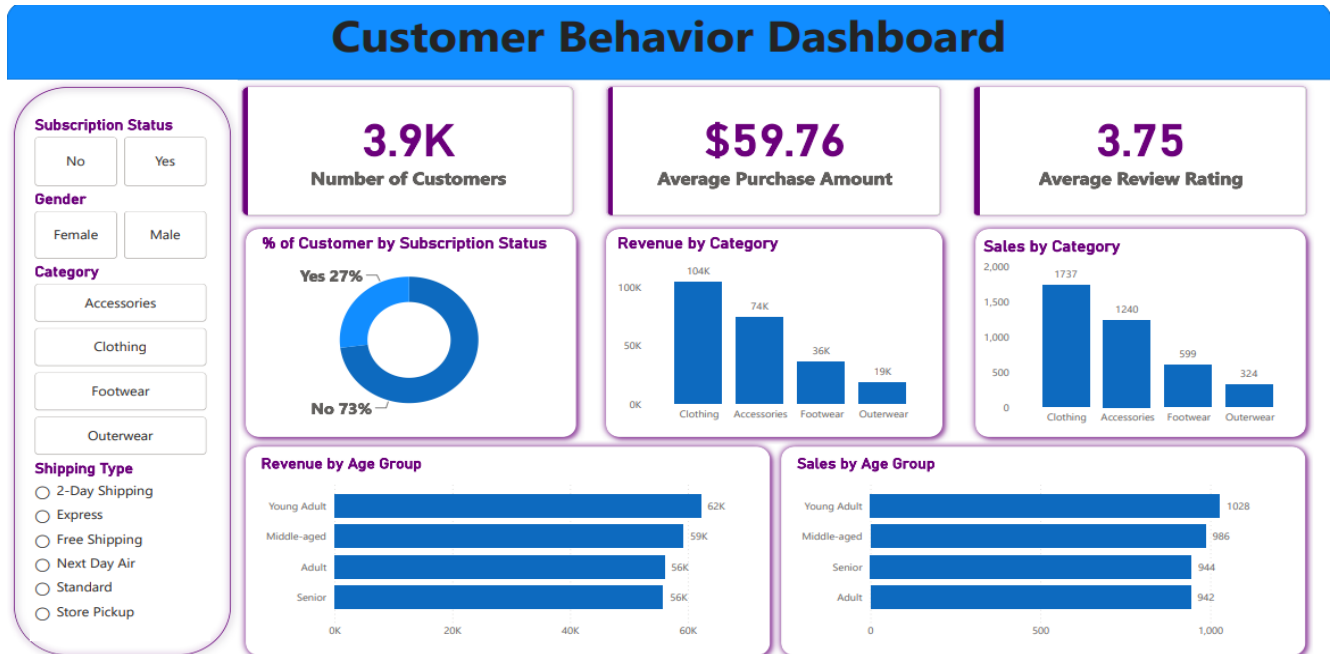
The dashboard allows users to explore customer behavior using filters and interactive charts.

Dashboard Overview

The dashboard provides a comprehensive overview of:

- Customer demographics
- Revenue performance

- Product sales
- Customer segments
- Subscription analysis



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.